

Cenxi Zhang

TOP 5% · AI AGENT SYSTEMS · EMBODIED DATA · ROBOTICS

Tel: +86 13561355595 | Email: cenxiapply@163.com | GitHub: github.com/zcxixixi

EDUCATION

Qingdao University of Technology

Sep 2023 - Jun 2027

B.Eng. in Building Electrical and Intelligent Engineering

GPA: 4.55 / 5.0 | Rank: 3 / 70 (Top 5%) | IELTS Academic: 6.5 | First-Class Scholarship for three consecutive years

POSITIONING

AI Agent & Embodied Intelligence Engineer who led a 360° video-to-UMI/VLA training-data engine, delivered three merged PRs to a 40k+ Star agent project, and led a team to a National First Prize in embedded vision and control.

RESEARCH INTERESTS

Agentic Systems; Vision-Language-Action Learning; Embodied Data and Evaluation; World Models; Long-Horizon Planning; Multimodal Human-Robot Interaction.

EXPERIENCE

Current Robotics

Jul 2026 - Present

Embodied AI & Robot Learning Systems Intern | UMI / VLA / 6DoF / Data Infrastructure

- Led the module from 0 to 1 for a 360° camera-to-UMI/VLA data engine, converting dual-fisheye video into multi-view RGB, dual-gripper 6DoF TCP trajectories, gripper states, validity masks, and UMI-compatible Zarr replay buffers.
- Built a hardware-aware localization and fusion stack integrating fisheye calibration, spherical multi-view AprilTag detection, IPPE/PnP, IMU disambiguation, optical-flow recovery, Kalman/RTS smoothing, and CAD/URDF extrinsics.
- Designed fail-closed quality gates and frame-level provenance for camera identity, calibration hashes, left/right role binding, synchronization, interpolation limits, and measured/recovered/predicted states.
- Established 29 regression-test modules and a 406-frame immutable dual-gripper baseline; an internal diagnostic sequence produced 230/230 valid poses with angular residual P95 of 0.50°.
- Completed 40 commits and approximately 18.6k lines of tracked Python engineering in six days, with GPU overfit smoke tests, WebGL review tooling, and versioned report/audit artifacts. Products I contributed to were involved in RMB 100M+ financing.

HKUDS nanobot Open-Source AI Agent Project

Feb 2026 - Apr 2026

Open-Source Contributor | Python / Agent Runtime / IMAP / SMTP / CLI / GitHub

- Delivered three merged PRs to the University of Hong Kong Data Intelligence Lab's 40k+ Star nanobot project.
- Implemented a consent-gated IMAP/SMTP agent channel and fixed CJK long-text input and arrow-key behavior, extending external communication while improving multilingual CLI reliability.

32-Hour Graduate AI Project Course

Apr 2026 - Jul 2026

Course Designer & Lead Instructor | Neural Networks / CNNs / Transformers / LLM Agents / Generative AI

- Designed and taught an eight-unit, 32-hour project course for 93 master's students, progressing from mathematical foundations to working AI systems and deployment.
- Built reusable notebooks, datasets, automated assessment, and interactive demos; open-sourced materials through github.com/zcxixixi/ai-course and bilibili.com/video/BV1Nd596vE5e.

Embedded Vision Measurement and Control System

Aug 2025

Team Leader | National Undergraduate Electronics Design Contest | C / Python / YOLO / OpenCV

- Led a three-person team from 0 to 1 through a four-day, three-night build of a perception-to-control system; won National First Prize, ranked 3rd in Shandong, and placed in the national top 1%.
- Integrated YOLO/OpenCV perception, camera calibration, perspective transformation, overlapping-target separation, coordinate mapping, and serial control into a real-time closed loop. Demo: youtube.be/vVcSWguqrLc.

LLM-Based Multimodal Wheel-Legged Interactive Robot

Jun 2025

Core Developer | LLM / YOLOv8 / STM32 / LVGL / IoT | East China First Prize

- Integrated LLM interaction, gesture recognition, environmental sensing, IoT control, and wheel-legged actuation into a perception-decision-control-feedback loop.
- Owned STM32 motion control, sensor integration, and the LVGL human-robot interface for real-time status display and operator interaction. Demo: youtube.be/dhhkG9M1-zg.

SELECTED ROBOTICS PROJECTS

Friction-Aware ADRC-NN Control for Multi-DOF Manipulators

Sep 2025 - Present

Core Developer | MATLAB / Simulink / Python / ESO / Neural Control

- Designed a cooperative ADRC and neural-network controller for nonlinear low-speed manipulator friction, including the ESO-NN architecture, data cleaning, feature preparation, and simulation validation.
- Built PID and ADRC baselines and compared tracking behavior, robustness, and low-speed control performance without overstating simulation results as hardware deployment.
- Used simulation trajectories to analyze friction-sensitive motion regimes and isolate the contribution of disturbance estimation and neural compensation.
- Structured the study around reproducible baselines, feature preparation, and ablation-ready evaluation rather than a single best-case trajectory.

TECHNICAL LEADERSHIP

Innovation Laboratory, School of Information and Control Engineering

Jan 2024 - Present

Student Technical Lead | Qingdao University of Technology

- Lead technical training, project mentoring, and competition preparation across LLM agents, computer vision, embedded systems, and control.
- Mentor undergraduates through OpenCV, PID, Kalman filtering, STM32 integration, and real-time debugging, turning algorithm design and engineering retrospectives into repeatable project workflows.
- Translate research concepts into buildable milestones, review interfaces between perception and control, and guide teams through failure analysis before competition delivery.

Selected Embedded and Robotic Systems

Sep 2024 - Jun 2025

Core Developer | STM32 / Control / Computer Vision / Robotic Actuation

- Wind Pendulum:** developed dual-axis PID with feedforward compensation for disturbance-resistant oscillation control and target tracking. Demo: youtu.be/A5QPHtyPR2g.
- High-Precision Line-Following Vehicle:** combined visual line detection, path tracking, and motor regulation for closed-loop navigation. Demo: youtu.be/UzqoWBQiwA0.
- Two-Wheel Self-Balancing Vehicle:** implemented Kalman attitude estimation, cascaded PID, and real-time motor regulation for autonomous balancing.
- Tic-Tac-Toe Robot:** integrated board-state vision, Minimax decision logic, and robotic-arm actuation into a complete human-machine gameplay loop.

REPRESENTATIVE CAPABILITY EVIDENCE

Agent Systems:	Shipped consent-gated IMAP/SMTP communication and CJK CLI fixes through three merged nanobot PRs; personal scope covered module implementation, compatibility fixes, and open-source delivery.
Embodied Data:	Connected 360° video, multi-view localization, dual-gripper 6DoF, UMI Zarr, and fail-closed quality gates; validated with 29 regression-test modules and a 406-frame immutable baseline.
Robot Systems:	Delivered two perception-decision-control competition systems as team leader and core developer, resulting in National First Prize and East China First Prize.

HONORS AND AWARDS

- National First Prize**, National Undergraduate Electronics Design Contest, 2025.
- First Prize, East China Division**, National Embedded Chip and System Design Competition, 2025.
- First-Class Scholarship for Outstanding Students, Qingdao University of Technology, 2024, 2025, and 2026.
- Third Prize, Lanqiao Cup National Finals, Embedded Design and Development, 2025.
- First Prize, National Undergraduate Electronics Design Contest, Shandong Division, 2024.

SELECTED PUBLIC LINKS

GitHub:	github.com/zcxixixi
AI Course:	github.com/zcxixixi/ai-course bilibili.com/video/BV1Nd596vE5e
Robot Demos:	youtu.be/vVcSWguqrLc youtu.be/dhhkG9M1-zg youtu.be/A5QPHtyPR2g youtu.be/UzqoWBQiwA0

LANGUAGES

Chinese:	Native
English:	IELTS Academic 6.5